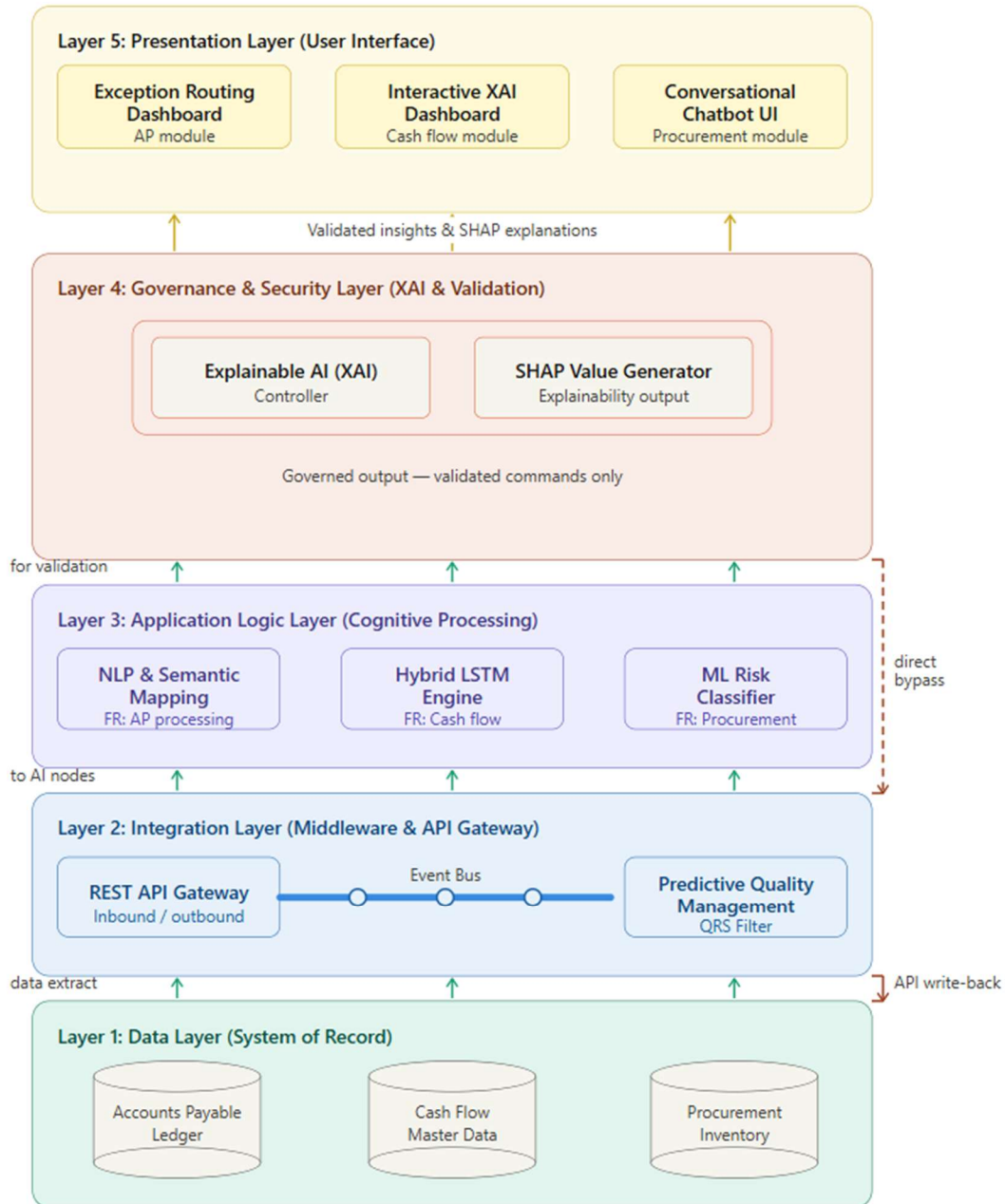


Track 1: AI for Accounting SW 1



[Table of Contents]

Project Features.....	3
1. AI-ERP Integration for Automated Financial Accounting.....	3
1.1. Business Process 1 - Automated Invoice Ingestion and Semantic Mapping.....	4
1.1.1. Use Cases	4
1.1.2. Sequence Diagram.....	5
1.1.3. User Stories	5
1.2. Business Process 2 - Cash Flow Forecasting and Liquidity Management.....	6
1.2.1. Use Cases	6
1.2.2. Sequence Diagram.....	7
1.2.3. User Stories	7
1.3. Business Process 3 - Interactive Decision Support and ML-Evaluated Procurement Risk Assessment.....	8
1.3.1. Use Cases	8
1.3.2. Sequence Diagram.....	9
1.3.3. User Stories	9
Field Requirements / Business Rules.....	10
Wireframe	12
Test Cases.....	12

Project Features

1. AI-ERP Integration for Automated Financial Accounting

The new feature brings a 5 layered AI integration architecture that covers 3 key modules of the Financial Accounting module: Accounts Payable (AP), Cash Flow Management, and Procurement. Traditional financial accounting processes are reactive, manual, and error-prone, requiring humans to enter data, apply rules, and generate reports, and are also slow, difficult to manage, and prone to inaccuracies given the increasing amount of unstructured financial data.

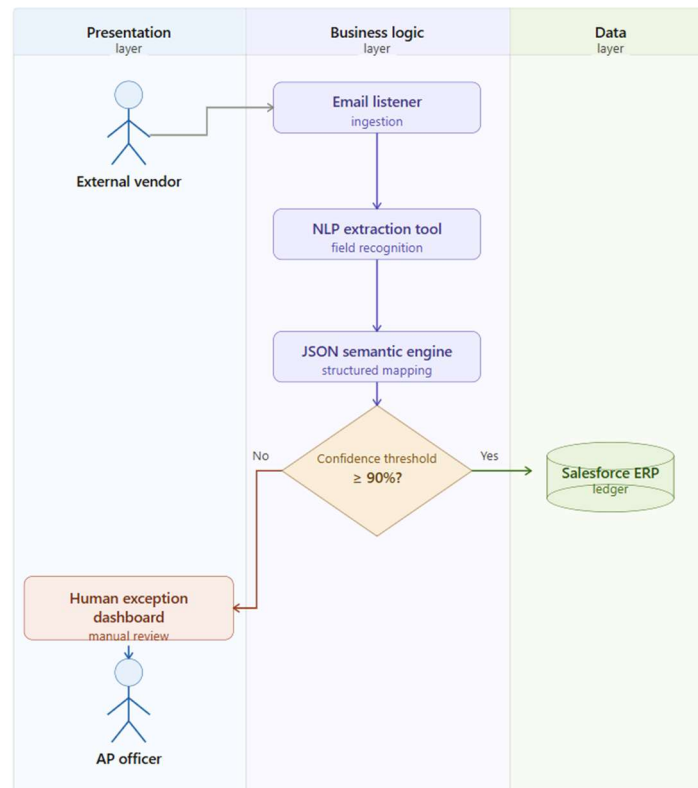
This is the gap that the proposed feature is meant to resolve: The AI middleware layer is an externalized application that is situated between the external business environment and the ERP system of record. The architecture separates all cognitive workloads into a controlled API gateway, avoiding the direct integration of non-deterministic AI into the fixed ERP system that would increase the risk of system instability, data integrity issues and regulatory compliance concerns. In the AP module, this gateway uses a Large Language Model, Google Gemini 2.5 Flash, to extract information from unstructured data; for time series cash flow forecasting, it integrates a Hybrid LSTM deep learning engine; for procurement intelligence, it's linked to a Retrieval-Augmented Generation (RAG) orchestrator and an ML Risk Classifier.

In all three modules, the same Governance and Security Layer ensures algorithmic transparency by using SHapley Additive exPlanations (SHAP) values and sends low confidence or high-risk outputs to Human-in-the-Loop (HITL) interfaces before any data is written back to the ERP ledger. This means that the master financial database never gets corrupted by unvalidated AI output, and finance workers are promoted to become strategic exception governors.

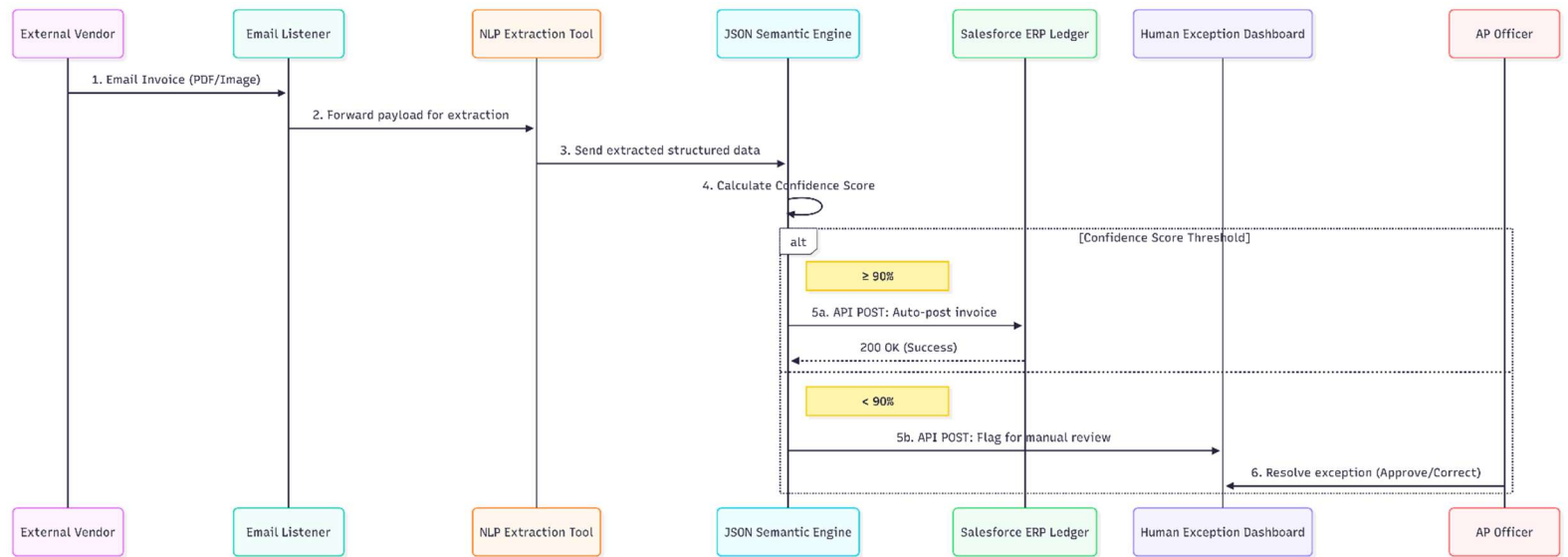
1.1.Business Process 1 - Automated Invoice Ingestion and Semantic Mapping

This is the primary automation workflow. It is triggered automatically every time an unread email arrives in the AP gateway inbox. The system processes both PDF invoice attachments and plain-text conversational invoice emails without any human initiation.

1.1.1. Use Cases



1.1.2. Sequence Diagram



1.1.3. User Stories

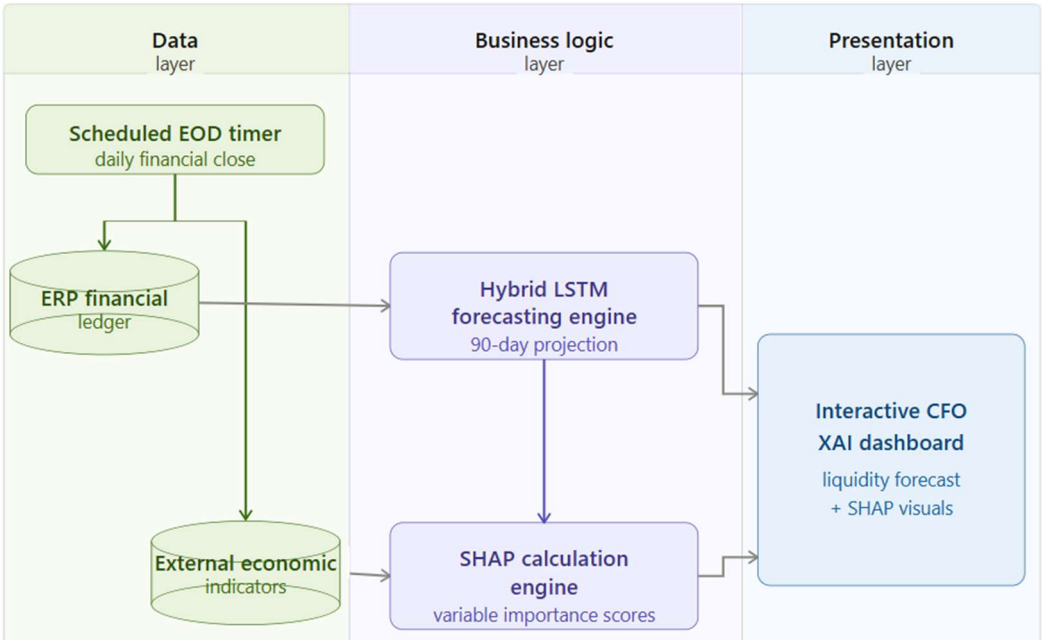
Requirement ID	Epic	User Story Summary
REQ-AP-1	Invoice Ingestion	As an Accounts Payable Officer, I want the system to automatically monitor the AP inbox for new vendor invoices so that I do not need to manually check and forward emails.
REQ-AP-2	Invoice Ingestion	As an Accounts Payable Officer, I want the AI to extract vendor name, invoice date, and total amount from both PDF and plain-text email invoices so that data entry is eliminated for standard invoices.
REQ-AP-3	Data Normalization	As an Accounts Payable Officer, I want all invoice dates to be automatically converted to ISO format (YYYY-MM-DD) and all amounts to be stored as clean floating-point numbers so that the ERP schema is never violated.
REQ-AP-4	Governance & Quality Control	As an Accounts Payable Officer, I want the system to assign a confidence score to every AI extraction and block low-confidence payloads from entering the master ledger so that data integrity is protected.
REQ-AP-5	Exception Handling	As an Accounts Payable Officer, I want low-confidence invoices to appear in an exception dashboard with an AI-generated warning reason so that I can quickly understand and resolve anomalies.

REQ-AP-6	Explainability	As an Accounts Payable Officer, I want to see a plain-language explanation of why an invoice was flagged ("Ambiguity in total amount") so that I can satisfy audit trail requirements without interpreting raw SHAP values.
REQ-AP-7	Human-in-the-Loop	As an Accounts Payable Officer, I want the exception dashboard to pre-populate AI-extracted fields (vendor name, date) even for flagged records so that I only need to correct the missing or ambiguous field rather than re-entering all data.
REQ-AP-8	Ledger Security	As a System Administrator, I want the AI processing workload to run in an externalized middleware layer (decoupled gateway) so that live ERP transactions are never delayed or blocked by AI inference calls.

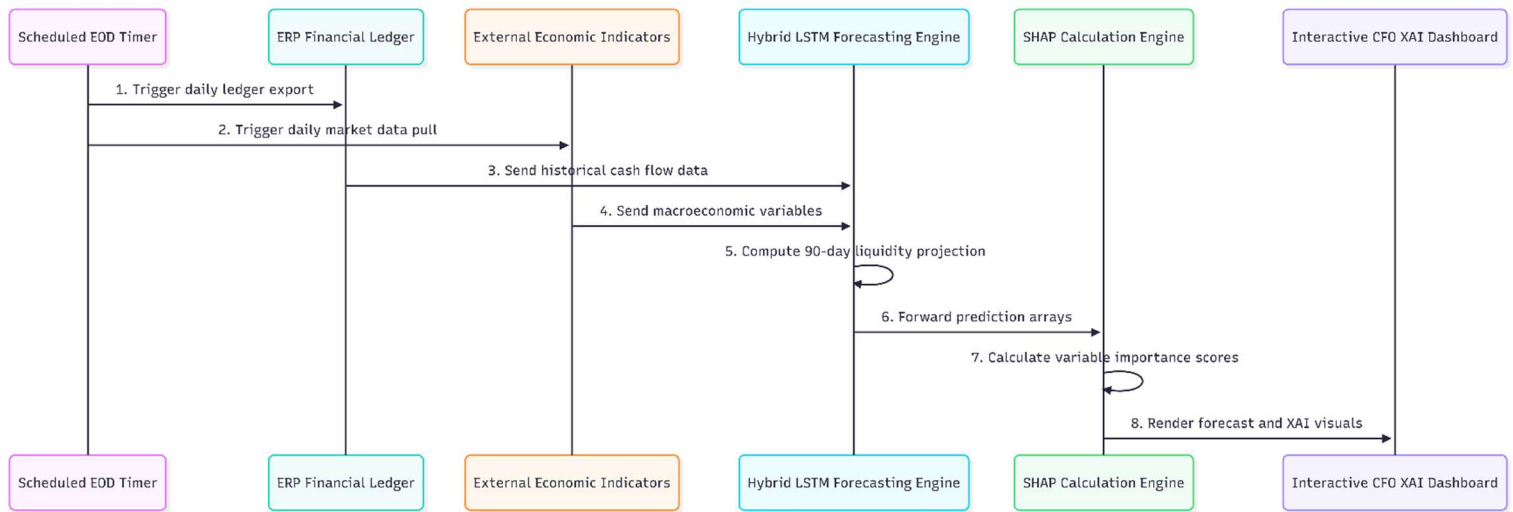
1.2.Business Process 2 - Cash Flow Forecasting and Liquidity Management

This workflow extends the AP automation to Cash Flow Management module. It is activated at the end of the day (daily ledger closure) according to a set time or when prompted by an ad-hoc CFO query. System fetches the historical transaction data from ERP systems and the real-time macroeconomic data, feeds it to a Hybrid LSTM forecasting engine, passes the forecast through the SHAP governance layer, and displays the explainable forecast on an Interactive XAI Dashboard.

1.2.1. Use Cases



1.2.2. Sequence Diagram



1.2.3. User Stories

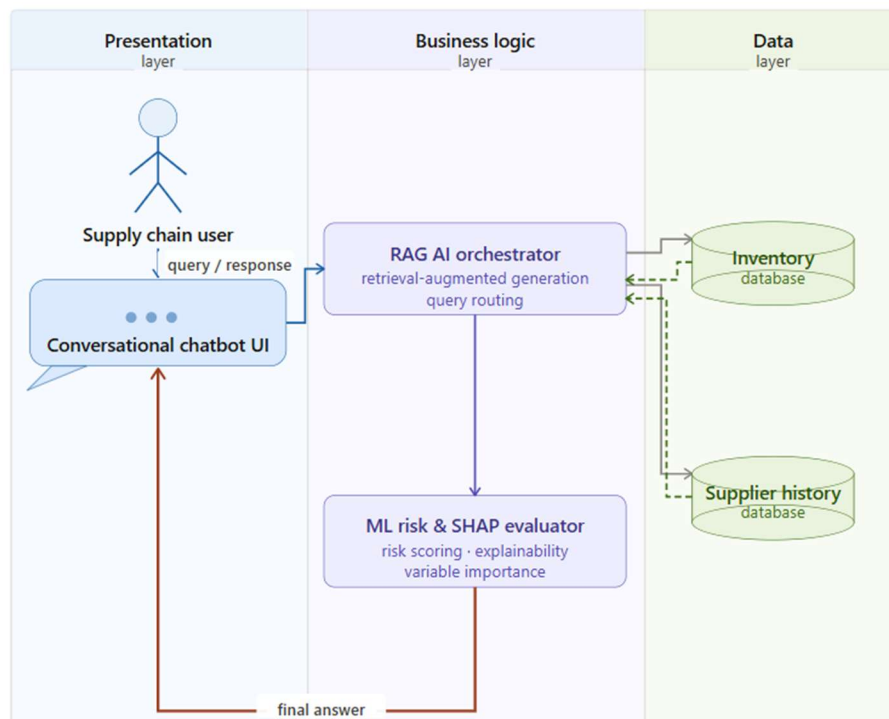
Requirement ID	Epic	User Story Summary
REQ-CF-1	Automated Forecasting	As a CFO, I want the system to automatically generate a 90-day cash flow forecast every evening after ledger close so that I have updated liquidity projections available every morning.
REQ-CF-2	Data Integration	As a Finance Analyst, I want the forecasting engine to combine internal ERP transaction data with real-time macroeconomic indicators (inflation, interest rates) so that projections reflect both internal and external financial conditions.
REQ-CF-3	Forecast Accuracy	As a CFO, I want the system to use a Hybrid LSTM model rather than static ARIMA averages so that non-linear, complex cash flow dependencies are captured more accurately.
REQ-CF-4	Explainability	As a Finance Auditor, I want each forecast to be accompanied by SHAP variable importance scores that identify the top drivers of the projected cash position so that I can explain forecast assumptions to the board.
REQ-CF-5	Ad-hoc Queries	As a Finance Manager, I want to trigger an off-cycle cash flow forecast at any time through the XAI dashboard so that I can respond to sudden market events without waiting for the scheduled daily cycle.

REQ-CF-6	Governance	As a Chief Compliance Officer, I want all AI-generated cash flow projections to pass through the XAI governance layer before being presented to decision-makers so that no unvalidated or black-box output is ever acted upon.
----------	------------	--

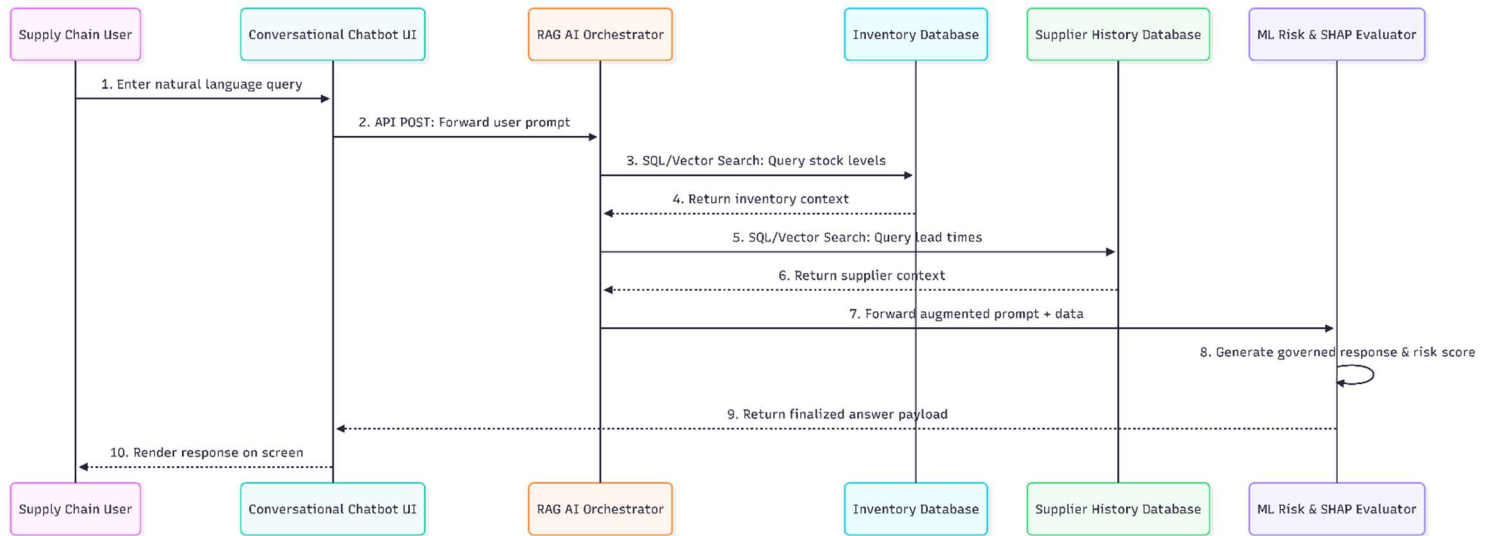
1.3.Business Process 3 - Interactive Decision Support and ML-Evaluated Procurement Risk Assessment

It is a completely self-initiated and user-driven process. A supply chain user enters a natural language question into a Conversational Chatbot UI, such as "What is the stock level of Component X and is Supplier Y reliable?"). The RAG AI Orchestrator is the engine that receives the query, pulls in real-time data from the Inventory Database and Supplier History Database, passes the enhanced prompt to an ML Risk and SHAP Evaluator, which scores the risk and returns a governed answer, with explanation to the user. By using the architecture, the AI will not hallucinate, as it will always refer to the current data within the enterprise, and each output will include a risk score and a breakdown of the SHAP variables, allowing for transparency and understanding.

1.3.1. Use Cases



1.3.2. Sequence Diagram



1.3.3. User Stories

Requirement ID	Epic	User Story Summary
REQ-PR-1	Conversational Query Interface	As a Supply Chain Manager, I want to ask procurement questions in plain natural language through a chatbot interface so that I do not need to manually run ERP reports or write SQL queries to get inventory and supplier insights.
REQ-PR-2	Inventory Retrieval	As a Supply Chain Analyst, I want the system to automatically retrieve real-time stock levels and lead times from the Inventory Database when I ask a stock-related question so that my decisions are always based on current ERP data rather than cached reports.
REQ-PR-3	Supplier Risk Assessment	As a Procurement Officer, I want the system to retrieve a supplier's historical delivery performance and risk profile from the Supplier History Database when I ask a supplier-related question so that I can make evidence-based sourcing decisions.
REQ-PR-4	Hallucination Prevention	As a CFO, I want every AI-generated procurement response to be grounded exclusively in live data retrieved from the ERP's Inventory and Supplier History databases so that the system never fabricates stock figures or supplier ratings.
REQ-PR-5	ML Risk Scoring	As a Supply Chain Manager, I want each procurement response to include a machine learning risk classification (Low / Medium / High risk) for the

		supplier or inventory situation described so that I can quickly prioritize which issues require immediate action.
REQ-PR-6	SHAP Explainability	As a Chief Procurement Officer, I want to see the top SHAP variable importance drivers behind each risk score (e.g., "Late delivery rate: 38% contribution", "Average lead time variance: 27% contribution") so that I understand why a supplier is classified as high-risk and can explain it to stakeholders.
REQ-PR-7	Governance Before Display	As a Compliance Officer, I want every AI procurement response to pass through the ML Risk and SHAP Evaluator before being rendered on the chatbot screen so that no unvalidated or unexplained AI output is ever shown to a decision-maker.
REQ-PR-8	Multi-Class Risk Profiling	As a Supply Chain Analyst, I want the system to classify supplier delivery performance into at least three categories (Ahead, On-Time, Late) rather than a binary flag so that I have a more nuanced view of supply chain resilience as established in the literature (Kang & Bhawna, 2025).
REQ-PR-9	Contract Gap Detection	As a Procurement Officer, I want the system to flag missing or non-standard clauses in supplier contracts when I upload them for review so that legal and compliance risks are identified before contracts are signed (Zhou et al., 2025).

Field Requirements / Business Rules

Vendor_Invoice__c - Accounts Payable Ledger Object (Salesforce):

Field Name	Data Type	Business Rule
Vendor_Name__c	Text (100)	Required. Must be a non-empty alphanumeric string.
Invoice_Date__c	<i>Date</i>	Required. Must be in ISO format YYYY-MM-DD. Any natural language date ("January 25th") or regional format (DD/MM/YYYY) will cause a schema validation error. The AI gateway must normalize all dates before write-back.

Total_Amount__c	Currency (16, 2)	Required. Must be a clean floating-point number with no currency symbols (EGP, \$, EUR), no comma separators, and no ambiguous values (subtotals, late fees). The AI must extract only the final payable total.
Processing_Status__c	Picklist	Values: "AI Approved", "Pending Review", "Human Approved", "Rejected". Defaults to "AI Approved" on auto-post; "Pending Review" on exception routing.

Invoice_Exception__c - Quarantine Object (Salesforce):

Field Name	Data Type	Business Rule
Vendor_Name__c	Text (100)	Pre-populated by AI (may be partial or "Unknown"). Editable by AP Officer.
Invoice_Number__c	Text (50)	Pre-populated by AI if detected; otherwise, blank for manual entry.
Extracted_Total__c	Currency (16, 2)	Pre-populated by AI (may be 0.0 if missing). AP Officer must correct before approval.
AI_Confidence_Score__c	Number (3, 0)	Read-only. Value that triggered the exception (always < 90).
AI_Warning_Reason__c	Long Text (500)	Read-only. Plain-language explanation from the AI (e.g., "Ambiguity in total amount: two values present, sender is unsure which to pay.").
Status__c	Picklist	Values: "Pending Review", "Human Approved", "Rejected".

Business Rules:

BR-01: No record may be written to Vendor_Invoice__c without a confidence score ≥ 90 . The API gateway enforces this check before every Salesforce API call.

BR-02: A record in Invoice_Exception__c may only be transferred to Vendor_Invoice__c after a human AP Officer sets Status__c to "Human Approved" via the Exception Dashboard flow.

BR-03: Spam or non-financial emails that produce a warning reason of "No invoice content detected" must be set to "Rejected" and deleted from Invoice_Exception__c after review.

BR-04: All date values entering Vendor_Invoice__c must conform strictly to ISO (YYYY-MM-DD). The gateway prompt explicitly instructs the AI to reformat all regional date formats before returning the JSON payload.

BR-05: All amount values entering Vendor_Invoice__c must be stripped of currency symbols and comma separators. The AI is instructed to return a pure float representing only the final total amount due.

BR-06: After every successful or failed invoice processing attempt, the email in the gateway inbox is marked as SEEN (read) to prevent duplicate processing.

BR-07: For security, any downloaded PDF file is deleted from the local gateway directory immediately after the Gemini API call completes.

Wireframe

<https://ap-exception-managem-kfzu.bolt.host>

Test Cases

Note: Test cases cover the Accounts Payable module only, as it is the sole module instantiated and empirically tested in this thesis. Cash Flow Management and Procurement were validated at the conceptual level through expert interviews only.

Test Case ID	Test Scenario	Test Steps	Expected Result
AP-TC-01	Process a standard PDF invoice with all required fields present	1) Send email with "Sliced Invoices" PDF attachment to gateway inbox. 2) Wait ≤ 10 seconds for IMAP poll cycle. 3) Check Salesforce Vendor_Invoice__c for new record.	New record created in Vendor_Invoice__c with Vendor_Name = "Sliced Invoices", Invoice_Date = "2016-01-25", Total_Amount = 93.5, Processing_Status = "AI Approved". Confidence score = 100.
AP-TC-02	Process an unstructured conversational email with distractors	1) Send email containing "Apex Office Supplies" invoice with European date (26/04/2026), 800 EUR subtotal, and waived 100 EUR delivery fee. 2) Wait for poll cycle. 3) Check Vendor_Invoice__c.	Record created with Vendor_Name = "Apex Office Supplies", Invoice_Date = "2026-04-26", Total_Amount = 845.5. Distractors (subtotal, waived fee) are ignored. Processing_Status = "AI Approved".
AP-TC-03	Governance filter intercepts incomplete invoice	1) Send email with a vendor invoice that omits the total amount. 2) Wait for poll cycle. 3) Check Invoice_Exception__c and AP Exception Dashboard.	No record written to Vendor_Invoice__c. Record created in Invoice_Exception__c with confidence score < 90 , Status = "Pending Review", AI_Warning_Reason populated. Record visible in AP Exception Dashboard.
AP-TC-04	Human resolution of exception record	1) Open AP Exception Dashboard. 2) Select an "Apex Office Supplies" exception with missing total. 3) Enter correct total amount. 4) Click "Approve & Post." 5) Check Vendor_Invoice__c.	Exception record's Status updates to "Human Approved." A new schema-compliant record appears in Vendor_Invoice__c with the corrected data. Exception record removed from pending queue.